07/03/2024

CMS-CST7XII_G

# Aakash

Medical | IIT-JEE | Foundations

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**MM : 720** CMS-003SE_CST-XII_Biology_2023-24_Test-4G **Time : 200 Min.**

**Topics Covered:**
**Complete Syllabus of 12th Class Zoology and Botany Subject.**

**General Instructions :**
**1. All questions are compulsory.**
2. Each question carries **+4 marks**. For every wrong response, **-1 mark** shall be ducted from the total score.
Unanswered/ unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

## BOTANY

**1.** Read the following statements and select the **correct** option.
**Statement A:** Pollen grains lose viability within 30 minutes of their release in members of Rosaceae, Leguminoseae and Solanaceae.
**Statement B:** Pollen grains can be stored in liquid nitrogen at –196° C for years.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

**2.** In a Mendelian monohybrid cross, selfing of $F_1$ heterozygous tall plant will produce.

(1) $2/4^{th}$ individuals of $F_2$ progeny as homozygous tall plants
(2) $2/4^{th}$ individuals of $F_2$ progeny as heterozygous tall plants
(3) $2/4^{th}$ individuals of $F_2$ progeny as homozygous dwarf plants
(4) $3/4^{th}$ individuals of $F_2$ progeny as homozygous tall plants

**3.** Characteristics of a flower are as
a. Nectaries absent
b. Flowers packed into inflorescence.
c. Well exposed stamens.
d. Light weighted, non-sticky pollens.
On the basis of above features, flower must be

(1) Hydrophilous
(2) Anemophilous
(3) Entomophilous
(4) Cleistogamous

**4.** Find the **correct** match w.r.t. seed.

(1) Seed coat – Represents integument of ovary
(2) Micropyle – Represents pore of ovary through which pollen tube enters
(3) Hilum – Represents the point at which body of ovule was attached with funicle
(4) Endosperm – Represents product of syngamy

**5.** In a clinical diagnosis a female 'A' was identified with absence of secondary sexual characters and has rudimentary ovaries. Which of the given statements can be **true** for the female 'A'?

(1) The chromosome complement of the female cannot be 44 + XO
(2) The female is suffering from a sex recessive linked disorder
(3) Mother of this female may be producing abnormal egg (22 + O) and father producing normal sperm (22 + X)
(4) This female may pass this disorder to her daughters

**6.** Select the **incorrect** statement w.r.t antibiotics.

(1) Used to treat deadly diseases like leprosy, diphtheria

(2) They are 'pro-life' with reference to disease causing organisms

(3) Produced by some microbes that can kill or retard growth of other microbes

(4) First antibiotic was discovered by Alexander Fleming

**7.** Which one is **correct** sequence of development of zygote into embryo?

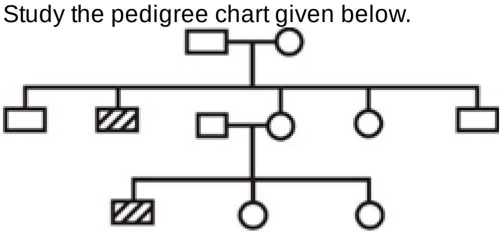
(1) Pro-embryo $\rightarrow$ Heart shaped embryo $\rightarrow$ globular embryo $\rightarrow$ Mature embryo

(2) Pro-embryo $\rightarrow$ Globular embryo $\rightarrow$ Heart shaped embryo $\rightarrow$ Mature embryo

(3) Heart shaped embryo $\rightarrow$ Pro-embryo $\rightarrow$ Globular embryo $\rightarrow$ Mature embryo

(4) Globular embryo $\rightarrow$ Heart shaped embryo $\rightarrow$ Pro-embryo $\rightarrow$ Mature embryo

**8.** Study the pedigree chart given below.



The trait traced in above pedigree chart is

(1) X-linked dominant

(2) Y-linked

(3) Autosomal dominant

(4) Autosomal recessive

**9.** Select the **correct** option w.r.t. sex linked Mendelian disorder

(1) Sickle cell anaemia

(2) Thalassemia

(3) Colour blindness

(4) Phenylketonuria

**10.** Two plants of garden pea with the genotype RrYy (Round, Yellow seeds) are crossed with each other. What is the probability for the offsprings to possess only one of the dominant phenotypes?

(1) $\frac{4}{16}$

(2) $\frac{5}{16}$

(3) $\frac{6}{16}$

(4) $\frac{3}{16}$

**11.** Shedding of pollen grains in 60% of angiosperms occurs at A celled stage.
Select the **correct** option for 'A'.

(1) One

(2) Two

(3) Three

(4) Four

**12.** Four genes a, b, c and d are present on a chromosome. Percentage of recombination between a and b = 20%; a and d = 2%, c and b = 8% and d and b = 18%. What will be the sequence of genes on the chromosome?

(1) adcb

(2) cdab

(3) dcab

(4) acbd

**13.** Select the **correct** set of parents those can produce child with blood group 'O'.
a. $I^Ai \times I^Ai$
b. $I^Bi \times I^AI^A$
c. $ii \times I^Ai$
d. $I^BI^B \times I^Bi$

(1) a and c

(2) b and d

(3) a and d

(4) b and c

**14.** **Assertion ( A) :** Embryo development precedes endosperm development.
**Reason ( R) :** Early stages of embryogeny are different in both monocots and dicots though seeds are similar.
In the light of the above statements, Select the **correct** option

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**15.** Genotype of a true breeding tall pea plant can be written as
(a) TT
(b) Tt
(c) tt

(1) (a) only

(2) (b) only

(3) (c) only

(4) Both (a) & (b)

**16.** Select the **odd** one w.r.t. polygenic inheritance.

(1) It demonstrates segregation and assortment of genes controlling quantitative traits

(2) Extreme phenotypes are rare and the intermediate ones are more frequent

(3) Dominant alleles do not show cumulative effect

(4) Common examples are skin colour in man and human height

**17.** The **correct** sequence of different parts of an ovule from outside to inside is

(1) Nucellus, Integument, Embryo sac

(2) Integument, Nucellus, Embryo sac

(3) Nucellus, Funicle, Chalaza

(4) Integument, Funicle, Nucellus

**18.** Active chemical in *Rauwolfia vomitoria* is

(1) Nicotine

(2) Taxol

(3) Azadirachtin

(4) Reserpine

**19.** Productivity is expressed in terms of energy as

(1) $g\ m^{-2}$

(2) $(g\ m^{-2})\ yr^{-1}$

(3) $kcal\ m^{-2}$

(4) $(kcal\ m^{-2})\ yr^{-1}$

**20.** Select the **correct** option to fill in the blanks w.r.t. Rivet popper hypothesis.

| | |
|---|---|
| Airplane | ___A___ |
| ___B___ | Species |

(1) A-Ecosystem, B-Rivets

(2) A-Key species, B-Rivets on the wing

(3) A-Ecosystem, B-Key species

(4) A-Ecosystem, B-Wings

**21.** Term biodiversity was popularised by

(1) Edward Wilson

(2) Lindman

(3) P. Ehrlich

(4) R. May

**22.** Tropics show rich biodiversity because,
(a) Tropical areas receive more solar energy
(b) Tropical environment is relatively more constant and predictable than temperate ones.
(c) Tropical environment is less seasonal than temperate environment.

(1) Only (a) is correct

(2) Only (b) is correct

(3) Only (c) is correct

(4) All (a), (b) and (c) are correct

**23.** Consider the following equation
$\log S = \log C + Z \log A$
In the given species-area relationship equation, C represents

(1) Species richness

(2) Y-intercept

(3) Area

(4) Regression coefficient

**24.** The most important cause driving animals and plants to extinction is

(1) Habitat loss and fragmentation

(2) Alien species invasions

(3) Over-exploitation

(4) Co-extinctions

**25.** Which of the following statements is **correct**?

(1) Passenger pigeon became extinct due to co-extinction

(2) Species diversity increases from poles to equator

(3) Organisms at higher trophic level in food chain are less susceptible to extinction

(4) At present, the tropical rain forest occupies 14% of earth's land area which was previously only 0.6%

**26.** The **correctly** matched population interaction is

| | **Species A** | **Species B** | **Name of interaction** |
|---|---|---|---|
| (1) | + | – | Mutualism |
| (2) | – | + | Competition |
| (3) | – | – | Predation |
| (4) | + | – | Parasitism |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**27.** Least productive biome is

(1) Tropical rain forest

(2) Temperate forest

(3) Desert

(4) Coniferous forest

**28.** Graphical representation of a young or growing population

(1) Is a bell shaped age pyramid

(2) Shows zero growth rate

(3) Shows fewer number of individuals in pre-reproductive group

(4) Is a triangular shaped age pyramid

**29.** Match the column I with column II and select the **correct** option.

| | **Column I** | | **Column II** |
|---|---|---|---|
| a. | Pleiotropy | (i) | Human height |
| b. | Polygenic inheritance | (ii) | Phenylketonuria |
| c. | Incomplete dominance | (iii) | Blood group 'AB' |
| d. | Co-dominance | (iv) | Gene for starch grain size in pea |

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(iii), c(iv), d(i)

**30.** In *lac* operon, *lac i* codes for

(1) Beta-galactosidase

(2) Permease

(3) RNA polymerase

(4) Repressor

**31.** There were 40 fishes last year in a lake and 16 new fishes have been added through reproduction taking the current population to 56. Calculate the birth rate.

(1) 0.49 fishes per month

(2) 0.4 offsprings per fish per year

(3) 0.28 offsprings per fish per year

(4) 0.74 offsprings per fish per year

**32.** A population is growing in a habitat which has limited resources. This population will show the sequence of growth phases

(1) Lag phase → Log phase → Crash

(2) Log phase → Lag Phase → Asymptote

(3) Lag phase → Deceleration → Log phase → Acceleration

(4) Lag phase → Acceleration → Deceleration-→ Asymptote

**33.** Large holed swiss cheese is ripened by a

(1) Fungus

(2) Virus

(3) Protist

(4) Bacterium

**34.** UTRs (untranslated regions) are some additional sequences present on mRNA which **do not** undergo translation. These sequences are present at

(1) 5′ end only (before start codon)

(2) 5′ end (after start codon) and 3′ end (before stop codon)

(3) 5′ end (before start codon) and 3′ end (after stop codon)

(4) 3′ end only (after start codon)

**35.** **Statement A:** A daughter will not normally be colour-blind, unless her mother is atleast a carrier and father is colour-blind.
**Statement B:** A colour-blind man cannot pass the disorder to his sons but can pass on to grandsons.
Read the above statements and select the **correct** option.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A & B are incorrect

(4) Both statements A & B are correct

**36.** The condition that results in polyploidy is

(1) Segregation of chromosomes without DNA replication

(2) Unequal distribution of chromosomes during cell division

(3) Failure in cell division after DNA replication

(4) Presence of more than one origins of replication in DNA

**37.** Sex of the chicks is determined by

(1) Their female parent

(2) Both of their male and female parents

(3) The temperature at which the egg is incubated

(4) Their male parent

**38.** Genes in eukaryotes
(a) Have introns
(b) Are split
(c) Are mostly polycistronic

(1) (a) & (b) only

(2) (b) & (c) only

(3) (a) & (c) only

(4) All (a), (b) & (c)

**39.** Which of the following chromosome complements is **correct** for Klinefelter's Syndrome?

(1) 47, XXXX

(2) 45, XO

(3) 47, XXY

(4) 46, XYY

**40.** Which of the given nitrogenous bases are found in both DNA and RNA?
(A) Uracil
(B) Adenine
(C) Guanine
(D) Thymine

(1) A and B

(2) B and C

(3) C and D

(4) A and D

**41.** Primary productivity depends on all, **except**

(1) Availability of solar radiations

(2) Availability of nutrients

(3) Variety of carnivores present in an area

(4) Photosynthetic capacity of plants

**42.** Microbial biocontrol agents that can be introduced in order to control butterfly caterpillars is

(1) *Trichoderma*

(2) *Bacillus thuringiensis*

(3) *Oscillatoria*

(4) *Methanobacterium*

**43.** Find the **odd** one w.r.t. the anther wall layers.

(1) Epidermis

(2) Middle layers

(3) Aleurone layer

(4) Tapetum

**44.** Activated sludge contains

(1) Aerobic bacteria only

(2) Flocs

(3) Anaerobic bacteria only

(4) Anaerobic fungi and bacteria only

**45.** The last step of DNA fingerprinting is

(1) Blotting

(2) Hybridisation

(3) Electrophoresis

(4) Autoradiography

**46.** DNA is preferred as a genetic material over RNA because it

(1) Is structurally more stable

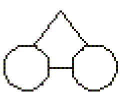
(2) Mutates faster

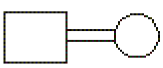
(3) Lacks replication

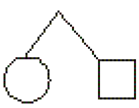
(4) Cannot express itself in the form of Mendelian characters

**47.** During pedigree analysis, consanguineous mating is represented by the symbol

(1) □

(2) 

(3) 

(4) 

**48.** Backbone of the polynucleotides chain is formed by

(1) Sugar and phosphates

(2) Only sugar

(3) Nitrogenous base and sugar

(4) Nitrogenous base and phosphates

**49.** During experiment with bacteriophage, Alfred Hershey and Martha Chase (1952) used

(1) $P^{32}$ labelled protein coat and $S^{35}$ labelled DNA

(2) $P^{32}$ labelled DNA and $S^{35}$ labelled protein coat

(3) $P^{32}$ labelled bacteria and $S^{35}$ labelled bacteriophage

(4) $P^{32}$ labelled bacteriophage and $S^{35}$ labelled bacteria

**50.** Select the **incorrect** match w.r.t. population interactions.

| | **Species A** | **Species B** | **Interaction** |
|---|---|---|---|
| (1) | Cattle egrets | Cattles | (+, 0) |
| (2) | Sparrow | Seeds | (+, –) |
| (3) | *Penicillium* secreting Penicillin | Bacteria | (+, –) |
| (4) | *Cuscuta* | Hedge plant | (+, –) |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**51.** Mark the **incorrect** statement w.r.t. features of human genome.

(1) The human genome contains 3164.7 million bp

(2) The average gene consists of 3000 bases

(3) Less than 2 percent of genome codes for proteins

(4) Chromosome 1 has fewest genes

**52.** RNA polymerase I transcribes all types of rRNAs **except**

(1) 28S rRNA

(2) 18S rRNA

(3) 5.8S rRNA

(4) 5S rRNA

**53.** _____ serve as an important biofertilizer in paddy fields.

(1) Fungi

(2) Mycoplasma

(3) Lichens

(4) Cyanobacteria

**54.** In which of the given organisms, Taylor *et.al*. have proved semiconservative mode of chromosome replication by using radioactive thymidine?

(1) *Vicia faba*

(2) *E. coli*

(3) *Pneumococcus*

(4) *Streptococcus*

**55.** In an operon model in bacteria, which of the given gene in *lac*-operon is a constitutive gene?

(1) Regulator gene

(2) z-gene

(3) y-gene

(4) Structural gene

**56.** How many maximum number of amino acids can be polymerised during translation from the given sequence of bases on mRNA if, 7th and 12th bases from 5′ undergo deletion?
5′AUGCACGAUGACUAUGA3′

(1) 3

(2) 4

(3) 5

(4) 2

**57.** The mathematical expression for the most realistic growth curve is

(1) $\frac{dN}{dt} = \mathrm{rN}$

(2) $\frac{\mathrm{dN}}{\mathrm{dt}} = \mathrm{rN}\ \left(\frac{\mathrm{K-N}}{\mathrm{K}}\right)$

(3) $\frac{\mathrm{B}}{\mathrm{D}} \times 100$

(4) $\frac{\mathrm{dN}}{\mathrm{dt}} = \left(\mathrm{b} - \mathrm{d}\right) \times \mathrm{N}$

**58.** Male honeybees

(1) Are diploid organisms

(2) Produce gametes by mitosis

(3) Are developed by fusion of male gametes with egg

(4) Do not have fathers but can produce sons

**59.** Which of the following adaptations ensures seed formation even in the absence of any pollinating agent?

(1) Protogyny

(2) Dichogamy

(3) Cleistogamy

(4) Protandry

**60.** According to 'Law of segregation'

(1) Two factors of a character get mixed up in $F_2$ generation

(2) During gamete formation, factors of a character get mixed up

(3) Factors of a character separate from each other during gamete formation

(4) Out of two factors for each character, dominant will be able to express in $F_1$ generation

**61.** Match column I with column II and select the **correct** option.

| Column I | Column II |
|---|---|
| a. Peptidyl Transferase | (i) Joins DNA fragments |
| b. DNA ligase | (ii) Activation of amino acid |
| c. Amino acyl tRNA synthetase | (iii) Unwinding of DNA helix |
| d. Helicase | (iv) Peptide bond formation |

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iii), b(i), c(iv), d(ii)

**62.** Meselson and Stahl experimentally proved semiconservative mode of DNA replication in *E. coli* using

(1) $^{15}N$ and $N^{14}$ isotopes

(2) Both $^{15}N$ and $S^{35}$ isotopes

(3) $^{14}N$ isotopes only

(4) Both $^{15}N$ and $P^{32}$ isotopes

**63.** All of the given are the components of transcription unit, **except**

(1) Promoter

(2) Regulator

(3) Structural gene

(4) Terminator

**64.** Which among the following play important role in guiding the pollen tube into the embryo sac?

(1) Synergids

(2) Antipodals

(3) Central cell

(4) Egg cell

**65.** Which of the following traits is present in wild-type *Drosophila*?

(1) Miniature wings

(2) Yellow body

(3) Red eyes

(4) White eyes

**66.** Double fertilisation involves fusion of all of the following **except**.

(1) Egg

(2) Male gametes

(3) Polar nuclei

(4) Synergids

**67.** Which of the given propositions based on the observation of Erwin Chargaff is **correct** for a double stranded DNA?

(1) A + G = T + C

(2) G = T

(3) $\frac{\mathrm{A+C}}{\mathrm{G+T}} => 1$

(4) $\frac{A+T}{G+C} = 1$

**68.** Who was the first amongst these to demonstrate the scientific basis of inheritance and variation by conducting hybridisation experiments?

(1) Hugo de vries

(2) Gregor Johann Mendel

(3) Morgan

(4) Punnet

**69.** A typical dicot embryo does not have

(1) Epicotyl

(2) Embryonal axis

(3) Epiblast

(4) Hypocotyl

**70.** In monocot seed, the sheath that encloses radicle is

(1) Coleoptile

(2) Coleorhiza

(3) Epicotyl

(4) Scutellum

**71.** During transcription in eukaryotes

(1) RNA polymerase I synthesises mRNA

(2) RNA polymerase II synthesises rRNA

(3) RNA polymerase II synthesises hnRNA

(4) RNA polymerase III synthesises mRNA

**72.** Gause's principle is related to

(1) Competitive exclusion

(2) Population density

(3) Competitive co-existence

(4) Resource partitioning

**73.** During sewage treatment process, the primary treatment of sewage involves all, **except**

(1) Formation of primary sludge

(2) Use of heterotrophic fungi and bacteria

(3) Filtration of floating debris

(4) Removal of grit

**74.** In angiosperms, microsporogenesis is referred to

(1) Formation of pollen mother cell from megaspore

(2) Formation of pollen mother cell from microspore

(3) Germination of pollen grain on the stigma

(4) Formation of microspores by meiotic division

**75.** In an angiospermic ovule, central cell of the embryo sac, prior to the entry of pollen tube, contains

(1) A single haploid nucleus

(2) One diploid and one haploid nuclei

(3) Two haploid polar nuclei

(4) One haploid secondary nuclei

**76.** Gametes of threatened species can be preserved in viable and fertile conditions for long periods in/through

(1) Hot spots

(2) Cryopreservation

(3) Sacred grooves

(4) Zoological parks

**77.** Tribal people are integral component of which of the given biodiversity conservation strategies?

(1) Biosphere reserves

(2) Botanical gardens

(3) Tissue culture

(4) Zoological parks

**78.** Exploring molecular, genetic and species level diversity for products of economic importance is called

(1) Bioprospecting

(2) Biofortification

(3) Bioremediation

(4) Biomagnification

**79.** Which one is **odd** w.r.t. population attributes?

(1) Birth rate

(2) Sex ratio

(3) Death

(4) Per capita death

**80.** Based on the source of food, organisms occupy a specific place in food chain that is known as their

(1) Trophic level

(2) Strata

(3) Standing crop

(4) Standing state

**81.** Ecologically, standing crop means the amount of living material present in

(1) Trophic level of green plants only at a given time

(2) Same trophic levels of only consumers per unit area

(3) Different trophic levels at a given time per unit area

(4) Same trophic level of only primary consumers per unit area

**82.** Heterozygous round - yellow seeded pea plants were self-crossed and total 1600 seeds were collected. What will be the total number of seeds with both homozygous recessive traits?

(1) 600

(2) 100

(3) 300

(4) 900

**83.** Which of the given is used as immunosuppressive agent in organ-transplant patients?

(1) Statin

(2) Lipase

(3) Cyclosporin A

(4) Streptokinase

**84.** If UUU codes for phenylalanine in almost all of the organisms then it shows that

(1) Code is nearly universal

(2) Code is degenerate

(3) Code is ambiguous

(4) All organisms share same genetic informations

**85.** DNA with more G ≡ C than A = T has

(1) High melting areas

(2) Low melting areas

(3) Unequal number of sugar and phosphates

(4) Free 2'-OH group

**86.** Given below is the sequence of mRNA
5′ AUCGGCUACGA 3′
Identify the **correct** sequence of nucleotides in coding strand of double stranded DNA read in 3′ → 5′ direction.

(1) 5′ TAGCCGATGCT 3′

(2) 3′ AGCATCGGCTA 5′

(3) 3′ TAGCCGATGCT 5′

(4) 5′ AGCATCGGCTA 3′

**87.** How many types of gametes will be produced by the individual with genotype AaBBCcDDEe?

(1) 8

(2) 4

(3) 2

(4) 16

**88.** If coding strand of a gene has CATGAGCATA sequence, then what will be the corresponding sequence of the transcribed mRNA?

(1) C A U G A G C A U A

(2) G U A C U C G U A U

(3) G A U C U C G U A U

(4) C A T G A G C A T A

**89.** Which of the given represents monohybrid test cross?

(1) TT × TT

(2) Tt × TT

(3) Tt × tt

(4) tt × tt

**90.** Apomixis is a mechanism in which

(1) Nucellus or integuments never participate in embryo formation

(2) Seeds are produced without fertilization

(3) A diploid egg is fertilized by a male gamete

(4) Embryo always develops from haploid cells

## ZOOLOGY

**91.** All the given statements are correct regarding ZIFT, **except**

(1) Ova from wife / donor (female) and sperms from husband / donor (male) are collected

(2) Zygote is formed under simulated conditions in the laboratory

(3) The zygote with upto 8 blastomeres could be transferred into fallopian tube

(4) The zygote with more than 8 blastomeres stage could be transferred into fallopian tube

**92.** **Assertion (A) :** Hepatitis-B and genital warts are not completely curable if detected early and treated properly.
**Reason (R) :** These STDs are non-curable due to absence or less-significant symptoms in the early stages of infection.
In the light of above statements, select the **correct** option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**93.** **Assertion (A) :** Bacterial cells are made competent before transformation.
**Reason (R) :** DNA is a hydrophilic molecule and cannot pass through cell membrane.
In the light of above statements, select the **correct** option.

(1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**94.** RNAi, which is employed in making tobacco plant resistant to nematode, is essentially involved in preventing the process of

(1) Transcription

(2) Translation of the mRNA

(3) Replication of the DNA

(4) Extension of the DNA

**95.** Choose the **odd** one w.r.t. analogous structures.

(1) Forelimbs of man and cheetah

(2) Eyes of octopus and of mammals

(3) Flippers of penguins and dolphins

(4) Wings of butterfly and birds

**96.** The method of introducing alien DNA by directly injecting it into the nucleus of an animal cell is called

(1) Biolistics

(2) Gene gun

(3) Microinjection

(4) Heat shock method

**97.** Toxin present in Bt cotton in form of protoxin, gets converted into active toxin by

(1) The action of saliva of insect

(2) Alkaline pH in gut of insect

(3) Acidic pH in gut of insect

(4) Action of microorganisms in gut of insect

**98.** Match Column I with Column II and choose the **correct** answer.

| | **Column I** | | **Column II** |
|---|---|---|---|
| a. | Implantation | (i) | Mitotic divisions |
| b. | Seminal vesicles | (ii) | Group of cells that differentiates into germ layers |
| c. | Inner cell mass | (iii) | Secrete viscous fluid present in semen |
| d. | Cleavage | (iv) | Embedding of blastocyst in the endometrium |

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(i), c(iv), d(ii)

**99.** The protein encoded by gene *cryIAb* controls.

(1) Mosquitoes

(2) Beetles

(3) Corn borer

(4) Flies

**100.** During human embryonic development, by the end of which week is the body covered with fine hair, eye-lids are separated and eyelashes are formed?

(1) 4 weeks

(2) 12 weeks

(3) 14 weeks

(4) 24 weeks

**101.** A drug that adversely affects the cardiovascular system of the body is obtained from

(1) *Cannabis sativa*

(2) *Erythroxylum coca*

(3) *Atropa belladona*

(4) *Papaver somniferum*

**102.** The milk produced by Rosie, a GMO is known to be rich in which of the following proteins?

(1) Human insulin

(2) Human alpha lactalbumin

(3) β-galactosidase

(4) Alpha-1-antitrypsin

**103.** Select the **odd** one among the following w.r.t techniques that allow early diagnosis of diseases.

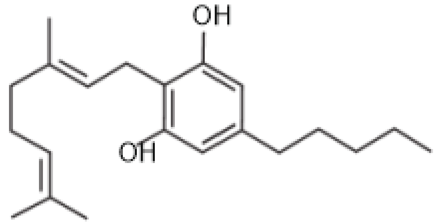
(1) ELISA

(2) PCR

(3) Western blot

(4) Serum analysis

**104.** A **false** statement about the following chemical structure is



(1) It is obtained from the inflorescence of the plant *Cannabis sativa*

(2) It affects the cardiovascular system of the body

(3) Is commonly called smack

(4) Its receptors are present principally in the brain

**105.** How many of the diseases given in the box below are vector borne?

| |
|---|
| Chikungunya, Filariasis, Dengue, Malaria, Pneumonia, Ascariasis, Diphtheria |

(1) Three

(2) Four

(3) Five

(4) Six

**106.** In simple stirred tank type of bioreactor, stirrer is responsible for

(1) Increase in the pH

(2) Decrease in temperature

(3) Extraction of product

(4) Uniform availability of oxygen

**107.** Where was Saheli developed?

(1) Central Drug Research Institute, Lucknow

(2) All India Institute of Medical Sciences, New Delhi

(3) National Institute of Immunology, New Delhi

(4) JNCASR, Bengaluru

**108.** Cave paintings by pre-historic humans can be seen at Bhimbetka rock shelter in Raisen district of

(1) Rajasthan

(2) Uttar Pradesh

(3) Madhya Pradesh

(4) Jharkhand

**109.** A gene locus has two alleles A, a. If the frequency of recessive allele a is 0.8, then what will be the frequency of, heterozygotes in a random mating population at equilibrium?

(1) 0.32

(2) 0.16

(3) 0.04

(4) 0.64

**110.** Which of the following proteins is used for treatment of emphysema?

(1) α-lactalbumin

(2) α-1-antitrypsin

(3) Tissue-plasminogen activator

(4) Interferon

**111.** According to 2011 census report, the world population was around

(1) 6 billion

(2) 6 million

(3) 7.2 billion

(4) 7.2 million

**112.** Assertion : The first clinical gene therapy was given in 1990 to a 4 year old girl with adenosine deaminase deficiency.

Reason : Gene therapy is used to treat hereditary diseases.

In the light of above statements, select the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

**113.** Steps of PCR do **not** include

(1) Annealing

(2) Denaturation

(3) Extension

(4) Transformation

**114.** If in a plasmid 'X' the sequence 5′ GAATTC 3′ appears at three sites, how many restriction fragments will result if we use *Eco*RI for restriction digestion?

(1) Two

(2) Three

(3) Four

(4) One

**115.** A divalent cation usually used to make the host cell competent is

(1) $Ca^{2+}$

(2) $Mg^{2+}$

(3) $Zn^{2+}$

(4) $Mn^{2+}$

**116.** Which of the following is not a critical research area of biotechnology?

(1) Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme

(2) Creating optimal conditions through engineering for a catalyst to act

(3) Use of better management practices and use of agrochemicals

(4) Downstream processing technologies to purify the protein/organic compound

**117.** A cellular defence method for silencing of a specific mRNA to prevent its translation is based on the process of

(1) Gene addition

(2) RNA interference

(3) Electrophoresis

(4) Elution

**118.** Read the following statements and choose the **correct** option.
**Statement A:** Oxytocin is known as milk let down hormone.
**Statement B:** It stimulates milk ejection in response to suckling reflex.

(1) Statement A is incorrect

(2) Statement B is incorrect

(3) Both statements A and B are incorrect

(4) Both statements A and B are correct

**119.** Complete the analogy w.r.t convergent evolution and select the **correct** option.
Bobcat : Tasmanian tigercat :: Lemur : ___________

(1) Numbat

(2) Mole

(3) Spotted cuscus

(4) Tasmanian wolf

**120.** For the transformation with recombinant DNA, cells are bombarded with high velocity microparticles of gold or tungsten, coated with DNA in a method known as

(1) Micro-injection

(2) Cloning

(3) Biolistics

(4) Elution

**121.** The genetic defect adenosine deaminase (ADA) deficiency may be cured permanently by

(1) Introducing ADA genes into bone marrow stem cells at early embryonic stage

(2) Enzyme replacement therapy

(3) Periodic infusion of genetically engineered lymphocytes having functional ADA cDNA

(4) Administering adenosine deaminase activators

**122.** Isolation of genetic material from fungal cells involves the use of all the following, **except**

(1) Chitinase

(2) Lysozyme

(3) Ribonuclease

(4) Protease

**123.** Sampling ports in a bioreactor are meant for

(1) Temperature control system

(2) pH control system

(3) Introduction of raw material

(4) Withdrawing small volumes of culture periodically

**124.** Interferons are secreted mainly by

(1) Natural killer cells

(2) All non-infected cells

(3) Plasma cells

(4) Virus infected cells

**125.** 'Panspermia' is favourite idea for some astronomers in context of origin of life. According to this concept

(1) Life was created by God.

(2) Life originated from non-living material like mud and straw.

(3) Life came on Earth as spores from some other planet.

(4) Life arose from some pre-existing life.

**126.** When the retrovirus enters macrophages, RNA genome of the virus replicates to form viral DNA with the help of

(1) Transcriptase inhibitors

(2) *Taq* polymerases

(3) Reverse transcriptase

(4) Proto-oncogenes

**127.** Hardy-Weinberg principle explains

(1) Size of population

(2) Average life span of population

(3) Genetic equilibrium

(4) Non-random mating

**128.** Liver cirrhosis is mostly caused by the chronic intake of

(1) Heroin

(2) Tobacco

(3) Morphine

(4) Alcohol

**129.** Insulin contains

(1) Two polypeptide chains only

(2) Three polypeptide chains only

(3) Four polypeptide chains only

(4) One polypeptide chain only

**130.** Golden rice has been developed for increased content of a substance which is considered as a precursor molecule for vitamin-A. This substance is

(1) Xanthophyll

(2) β-carotene

(3) Anthocyanin

(4) Retinol

**131.** A piece of double stranded DNA which was sent to the laboratory for amplification was put through 5 cycles of Polymerase Chain Reaction (PCR). How many copies will be obtained at the end of the process?

(1) 16

(2) 32

(3) 64

(4) 128

**132.** Read the following statements and select the **correct** option.
**I.** In blue/white colony selection, recombinants produce blue colonies due to insertional inactivation of α-galactosidase.
**II.** Transformants produced during genetic engineering include both recombinants and non-recombinants.

(1) Both statements are correct

(2) Only statement I is incorrect

(3) Only statement II is incorrect

(4) Both statements are incorrect

**133.** In genetic engineering, the most extensively used bacterium among the following is

(1) *Mycobacterium leprae*

(2) *Escherichia coli*

(3) *Neisseria gonorrhoeae*

(4) *Clostridium tetani*

**134.** Symptoms of ascariasis include all of the following, **except**

(1) Internal bleeding

(2) Muscular pain

(3) Intestinal perforation

(4) Anemia

**135.** *Hind* II was the first restriction endonuclease discovered. This enzyme cut the DNA molecules by recognising a specific sequence of

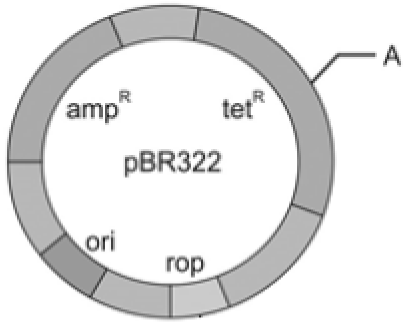
(1) 4 base pairs

(2) 2 base pairs

(3) 6 base pairs

(4) 8 base pairs

**136.** Restriction enzyme whose restriction site is represented by the region marked 'A' is




(1) *Pvu* I

(2) *Bam* HI

(3) *Pst* I

(4) *Hin*d III

**137.** Inactivation of gene in a plasmid due to joining in by any transgene is known as

(1) Insertional activation

(2) Insertional replication

(3) Insertional inactivation

(4) Transformation

**138.** If in a linear DNA, the recognition sequence for *Sma* I appears at 3 sites, then how many fragments will be produced?

(1) Two

(2) Three

(3) Five

(4) Four

**139.** 'Darwin's finches' present an example of all, **except**

(1) Differences in feeding habit

(2) Convergent evolution

(3) Founder's effect

(4) Adaptive radiation

**140.** Read the following points
a. Cranial capacity about 900 c.c.
b. Probably ate meat.
c. First prehistoric man to make use of fire.
These points are **true** for

(1) *Homo habilis*

(2) *Homo erectus*

(3) *Australopithecines*

(4) *Homo sapiens neanderthalensis*

**141.** Copper ions released from multiload 375 supress

(1) Sperm motility only

(2) Sperm motility and fertilising capacity of sperms

(3) Ovulation

(4) GnRH release from hypothalamus

**142.** Cancer cells show the property of __A__ while normal cells show property of __B__. Here A and B respectively represent

| | A | B |
|---|---|---|
| (1) | Contact inhibition | Metastasis |
| (2) | Rapid cell division | Escape from cell death |
| (3) | Metastasis | Contact inhibition |
| (4) | Contact inhibition | Programmed cell death |

Select the **correct** option.

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**143.** Ringworm is a fungal disease caused by
(A) *Microsporum*
(B) *Trichophyton*
(C) *Wuchereria*
(D) *Epidermophyton*
Choose the correct option

(1) (A) and (B) only

(2) (B) and (C) only

(3) (A), (B) and (C)

(4) (A), (B) and (D)

**144.** Which of the following **correctly** represents the sequence of evolutionary history of mammals?

(1) Early reptiles → Synapsids → Pelycosaurs → Therapsids → Mammals

(2) Extinct reptiles → Sauropsids → Pelycosaurs → Therapsids → Mammals

(3) Early reptiles → Pelycosaurs → Synapsids → Thecodonts → Mammals

(4) Early reptiles → Sauropsids → Thecodonts → Pelycosaurs → Mammals

**145.** Which of the following points are **correct** with reference to mutation according to the work of Hugo de Vries?
a. Large difference arising suddenly in a population.
b. Mutation causes evolution.
c. Mutations are random.
d. Single step large mutations are called saltation.
e. Mutations are directionless

(1) a, b & c only

(2) a, d & e only

(3) b, d & e only

(4) a, b, c, d & e

**146.** Which is the largest diverse group/unit that can show gene flow?

(1) Population

(2) Kingdom

(3) Phylum

(4) Genus

**147.** Select the **correct** option w.r.t. increasing cranial capacity suggested by hominid fossils

(1) *Homo erectus* → *Homo habilis* → *Homo sapiens sapiens* → *Homo sapiens fossilis*

(2) *Homo habilis* → *Homo erectus* → *Homo sapiens sapiens* → *Homo sapiens fossilis*

(3) *Homo erectus* → *Homo sapiens neanderthalensis* → *Homo habilis* → *Homo sapiens fossilis*

(4) *Homo habilis* → *Homo sapiens sapiens* → *Homo erectus* → *Homo sapiens fossilis*

**148.** *Latimeria* is considered as a connecting link between

(1) Fishes and amphibians

(2) Annelids and molluscs

(3) Cartilaginous fishes and bony fishes

(4) Annelids and arthropods

**149.** In humans, a complex neuroendocrine mechanism induces parturition and it involves

(1) Cortisol, FSH and LH

(2) Relaxin, oxytocin and prolactin

(3) Estrogen, cortisol and oxytocin

(4) Estrogen, progesterone and prolactin

**150.** The first human hormone produced by recombinant DNA technology is

(1) Estrogen

(2) Testosterone

(3) Thyroxine

(4) Insulin

**151.** Read the following statements A and B.
**Statement A** : MTPs are considered relatively safe during second trimester i.e. upto 12 weeks of pregnancy.
**Statement B** : The legal standard procedure of prenatal diagnosis to determine the sex of the unborn child is to perform amniocentesis followed by MTP in India.
Select the **correct** option.

(1) Statement A is correct

(2) Statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

**152.** A : Anti-venin given to patients of snake bites is an example of passive immunization.
R : Anti-venin is a preparation containing preformed antibodies against the snake venom.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**153.** The side-effects of the use of anabolic steroids in human females do **not** include

(1) Increased aggressiveness

(2) Abnormal menstrual cycles

(3) Breast enlargement

(4) Enlargement of clitoris

**154.** Which of the following animals are preferred for chemical safety testing and why?

(1) Non-transgenic animals; as they give the same result as transgenic animals

(2) Non-transgenic animals; as they are more sensitive to toxic substances

(3) Transgenic animals; as they are made more sensitive to toxic substances

(4) Transgenic animals; as they are immune to toxic substances

**155.** Who disproved the 'Good humor' hypothesis of health?

(1) Hippocrates

(2) William Harvey

(3) M.S Swaminathan

(4) Stanley Cohen and Herbert Boyer

**156.** Choose the **odd** one w.r.t. adaptive radiation in Australia.

(1) Wombat

(2) Bandicoot

(3) Sugar glider

(4) Flying squirrel

**157.** Who constructed recombinant DNA on the first instance?

(1) Har Gobind Khorana

(2) Hugo de Vries

(3) Kary Mullis

(4) Stanley Cohen and Herbert Boyer

**158.** Read the following statements carefully and choose the **correct** one from the given statements.

(1) DNA fragments can be visualised after electrophoresis, by exposure to UV radiation, without staining.

(2) DNA fragments can be visualised directly on the gel, without staining.

(3) After gel electrophoresis, the DNA fragments are stained using ethidium bromide followed by exposure to violet light of visible spectrum to visualise them.

(4) To visualise DNA fragments after gel electrophoresis, staining is done by ethidium bromide and then exposed to UV light.

**159.** Each restriction endonuclease functions by

(a) Inspecting the length of DNA sequence
(b) Binding to its specific recognition sequence
(c) Cut only single strand of dsDNA at specific points
(d) Cut in their sugar-nitrogenous base backbone

Choose the correct option from the following.

(1) a, b and c only

(2) a, b and d only

(3) a and b only

(4) a, b, c and d

**160.** The two enzymes responsible for restricting the growth of bacteriophage in *E.coli* were isolated in year

(1) 1963

(2) 1983

(3) 1990

(4) 1997

**161.** Which of the following is **not** a pair of secondary lymphoid organs in humans?

(1) Peyer patches, tonsils

(2) Lymph nodes, spleen

(3) Thymus, Bone marrow

(4) Tonsils, spleen

**162.** Test tube baby programme includes all, **except**

(1) ZIFT

(2) IUT

(3) GIFT

(4) ICSI

**163. Assertion (A) :** Amniocentesis is a foetal sex and genetic disorder determination test.

**Reason (R) :** Fluid secreted by foetus gives indication of sex and disorders in foetus.

In the light of above statements, select the **correct** option.

(1) Both Assertion & Reason are true and reason is the correct explanation of assertion

(2) Both Assertion & Reason are true but reason is not the correct explanation of assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**164.** Which era is recognised as age of reptiles?

(1) Cenozoic era

(2) Paleozoic era

(3) Mesozoic era

(4) Precambrian era

**165.** A test based on the chromosomal pattern in the cells of amniotic fluid surrounding the developing embryo is called amniocentesis. This test **cannot** determine

(1) Cleft palate

(2) Down's syndrome

(3) Haemophilia

(4) Sickle-cell anaemia

**166.** Which of the following factors has **not** contributed to population explosion in India since independence?

(1) Rise in MMR

(2) Decline in IMR

(3) Increase in number of people in reproductive age group

(4) Rapid decline in death rate

**167.** Acccording to Darwin, fitness of the individual is assessed in terms of

(1) Adaptive ability and reproductive fitness

(2) Strategies to obtain food and shelter

(3) Physical strength

(4) Number of mutations introduced in one generation

**168.** Drug that blocks the uptake of neurotransmitter dopamine is obtained from

(1) *Papaver somniferum*

(2) *Erythroxylum coca*

(3) *Cannabis sativa*

(4) *Atropa belladona*

**169.** Theory of spontaneous generation was disapproved by

(1) Oparin

(2) Louis Pasteur

(3) Haldane

(4) Cuvier

**170.** "It is a stochastic process based on chance events in nature and chance mutation in the organisms". This statement tells us about

(1) Non directional nature of evolution

(2) Directional mutations leading to natural selection

(3) Lack of anthropogenic impact on the evolutionary time scale

(4) Constant rate of reproduction & evolution in all organisms

**171.** Match following columns A and B.

| | **Column A** | | **Column B** |
|---|---|---|---|
| a. | Human heart is formed | (i) | During 5th month of gestation |
| b. | Development of limbs and digits | (ii) | At the end of first trimester of pregnancy |
| c. | Major organ systems are formed | (iii) | By the end of 2nd month of pregnancy |
| d. | First movement of foetus is observed | (iv) | After one month of pregnancy |

Choose the option containing all **correct** match.

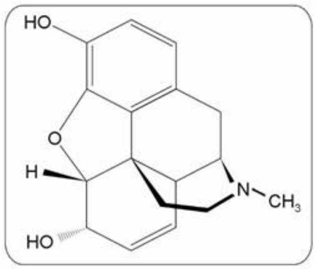
(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

**172.** Identify the drug represented by the chemical structure given below.



Select the **correct** option.

(1) Cocaine

(2) LSD

(3) Marijuana

(4) Morphine

**173.** Select the **mismatch** w.r.t. innate immunity.

| | | | |
|---|---|---|---|
| (1) | Physical barrier | – | Mucus coating of GIT |
| (2) | Physiological barrier | – | Acid in stomach |
| (3) | Cellular barrier | – | PMNL-neutrophils |
| (4) | Cytokine barrier | – | Natural killer cells |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

**174.** Which of the following diseases can be characterized by symptoms like constipation, abdominal pain, cramps, stools with excess mucous and blood clots?

(1) Amoebic dysentery

(2) Malaria

(3) Ringworm

(4) Typhoid

**175.** Given below are the statements w.r.t AIDS.

(A) HIV has an envelope enclosing the DNA genome.

(B) Treatment of AIDS with anti-retroviral drugs is only partially effective.

(C) AIDS was first reported in 1891 in India.

(D) A foetus can get the HIV infection from the infected mother through placenta.

Select the option which constitutes incorrect statements only.

(1) (A) and (B)

(2) (A) and (C)

(3) (C) and (D)

(4) (A), (C) and (D)

**176.** If nature favours those individuals in the population which possess the mean character value, then it is said that natural selection leads to

(1) Stabilisation

(2) Directional change

(3) Progressive change

(4) Disruption

**177.** Which of these options support Lamarckian view?

(1) Most geologic events occur slowly rather than catastrophically

(2) Only germ cells carry characters to offspring

(3) Development of organs is inversely proportion to its use

(4) Certain parts of the body get larger and complex through generations due to their extensive use

**178.** Match column I with column II w.r.t. female reproductive system in humans and select the **correct** option.

| | **Column I** | | **Column II** |
|---|---|---|---|
| (a) | Infundibulum | (i) | Membrane partially covering vagina |
| (b) | Stroma of ovary | (ii) | Tertiary follicle |
| (c) | Antrum | (iii) | Divided into cortex and medulla |
| (d) | Hymen | (iv) | Funnel shaped structure of oviduct |

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iii), b(ii), c(i), d(iv)

**179.** Which set of structures share similar ploidy levels?

(1) Spermatogonia, ovum

(2) Spermatid, secondary oocyte

(3) Ootid, primary oocyte

(4) Spermatozoa, primary spermatocyte

**180.** During parturition

(1) Oxytocin promotes relaxation of pubic symphysis

(2) Cervical mucus plug is further strengthened by estrogen and progesterone

(3) There is an increase in estrogen to progesterone ratio

(4) Prostaglandins are secreted from corpus luteum